

HOUSEHOLD ELECTRICITY CONSUMPTION EDA PROJECT

Exploratory Data Analysis using Python, Pandas, NumPy, Matplotlib, Seaborn and Streamlit

Project Type	End-to-End Exploratory Data Analysis
Dataset	Household Electricity Consumption
Records	1,044,506
Features	15
Analysis Focus	Consumption, electrical measurements and temporal patterns
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Streamlit

Project objective: understand the structure and behaviour of household electricity-consumption measurements, identify temporal patterns, examine distributions and provide data-driven observations that can support future analysis and machine-learning work.

Source basis: this report was prepared from the supplied Household analysis Jupyter Notebook and its saved statistical outputs and visualizations.

1. Project Overview

The supplied notebook presents an exploratory analysis of a household electricity-consumption dataset. The workflow begins with importing Python data-analysis libraries, loading cleaned data, checking dataset information and statistical summaries, and then examining numerical variables through distributions and temporal variables through grouped bar charts.

Why this project?

Electricity consumption varies over time and across electrical measurements. An exploratory analysis helps understand consumption levels, identify concentration and variation in the measurements, and observe how recorded values differ by year, month, day of the week and hour. These observations can later support feature engineering and predictive modelling.

Dataset Structure

No.	Feature	Description
1	Date	Recorded calendar date.
2	Time	Recorded time of observation.
3	Global_active_power	Global active power measurement.
4	Global_reactive_power	Global reactive power measurement.
5	Voltage	Voltage measurement.
6	Global_intensity	Global current intensity measurement.
7	Sub_metering_1	First sub-metering measurement.
8	Sub_metering_2	Second sub-metering measurement.
9	Sub_metering_3	Third sub-metering measurement.
10	Day_Name	Day of the week.
11	Month_Name	Month name.
12	Year	Calendar year.
13	Hours	Hour component of the observation time.
14	Minutes	Minute component of the observation time.
15	Total_metering	Combined sub-metering value used in the analysis.

2. Data Quality and Statistical Summary

The notebook's information output reports 1,044,506 observations and 15 columns. Every listed column has 1,044,506 non-null observations, so the saved notebook output shows no missing values in the analyzed dataset.

Measure	Result
Rows	1,044,506
Columns	15
Missing values shown by df.info()	0
Years represented	2006, 2007, 2008
Hour range	0–23
Minute range	0–59

Numerical Statistics

Feature	Mean	Std.	Min	Median	Max
Global_active_power	1.1082	1.1295	0.076	0.550	10.670
Global_reactive_power	0.1183	0.1095	0.000	0.098	1.390
Voltage	239.9598	3.2845	223.490	240.210	252.140
Global_intensity	4.7181	4.7636	0.200	2.400	46.400
Sub_metering_1	1.1775	6.3212	0.000	0.000	80.000
Sub_metering_2	1.4752	6.3527	0.000	0.000	78.000
Sub_metering_3	5.9338	8.2104	0.000	0.000	31.000
Year	2007.458	0.5389	2006	2007	2008
Hours	11.5046	6.9235	0	12	23
Minutes	29.4998	17.3177	0	30	59
Total_metering	8.5865	13.2558	0	1	129

The summary shows that Global_active_power, Global_reactive_power, Global_intensity and Total_metering are concentrated toward lower values while having substantially larger maximum values. Voltage is much more tightly concentrated around its central value.

3. Univariate Analysis

The notebook examines five numerical variables using histograms with KDE curves: Global_active_power, Global_reactive_power, Voltage, Global_intensity and Total_metering.

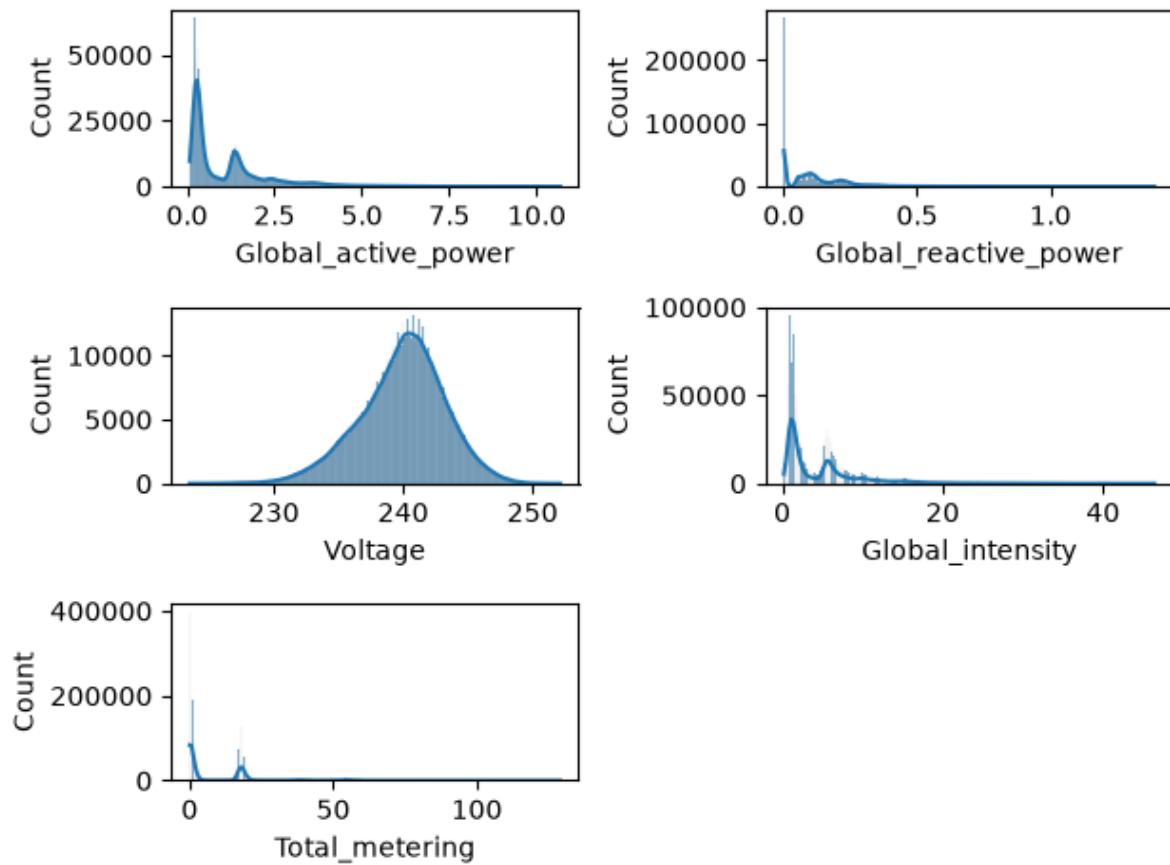


Figure 1. Numerical distributions reproduced from the supplied notebook output.

- **Global_active_power:** the distribution is strongly concentrated near lower values with a long right tail.
- **Global_reactive_power:** most observations are near the lower end of the scale, with a comparatively small number of larger observations.
- **Voltage:** the KDE is comparatively compact and centred around approximately 240 V.
- **Global_intensity:** the distribution is concentrated at low values and extends toward substantially higher intensities.
- **Total_metering:** values are heavily concentrated near the lower end, while the maximum reaches 129.

4. Temporal and Grouped Analysis

The notebook groups Total_metering, Global_active_power, Global_reactive_power and Voltage by Year, Month_Name, Day_Name and Hours. These charts provide a view of how the recorded measurements vary across time-related categories.

Total Metering by Time Categories

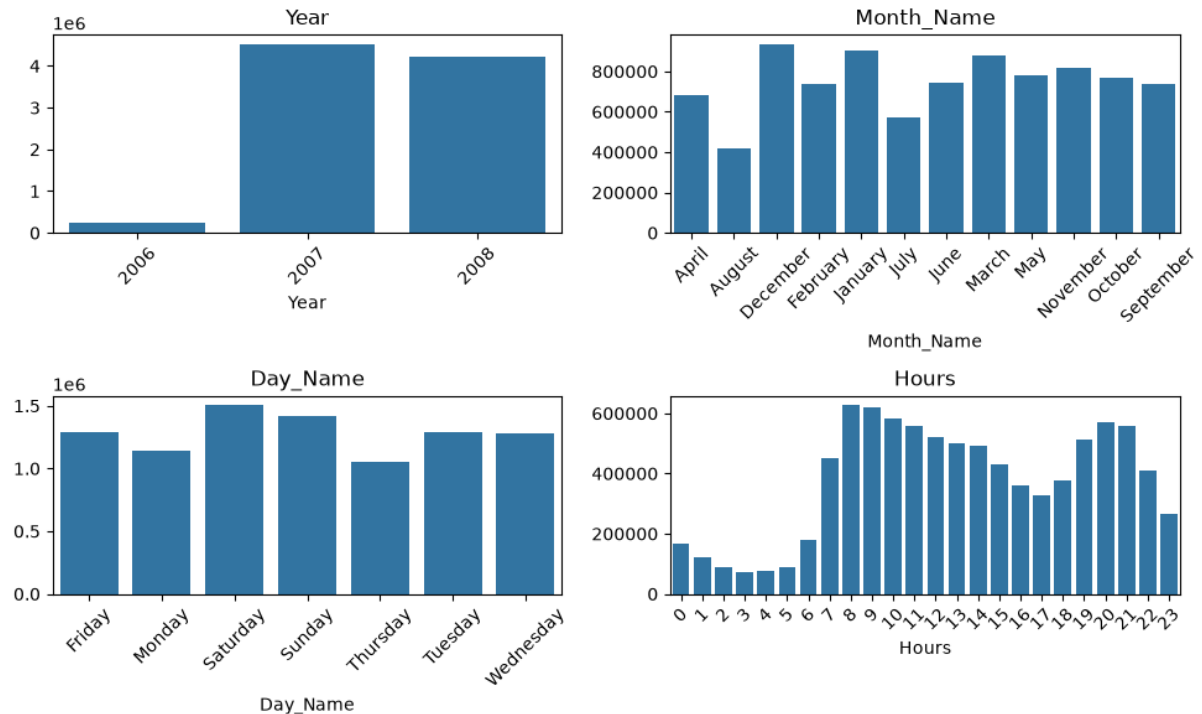


Figure 2. Total-metering grouped analysis from the notebook. The hourly distribution is visibly different across the day, with higher activity around the morning and evening periods and lower activity during several overnight/early-morning hours.

Global Active Power by Time Categories

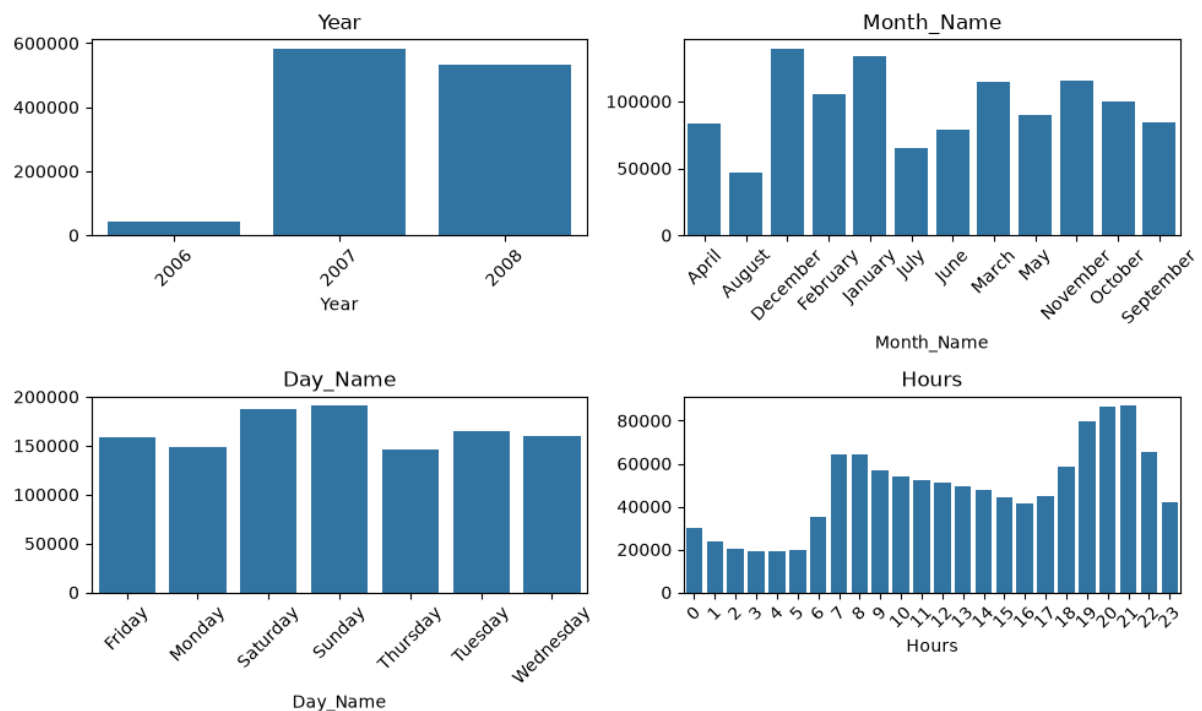


Figure 3. Global-active-power grouped analysis from the notebook.

5. Reactive Power and Voltage Patterns

Global Reactive Power

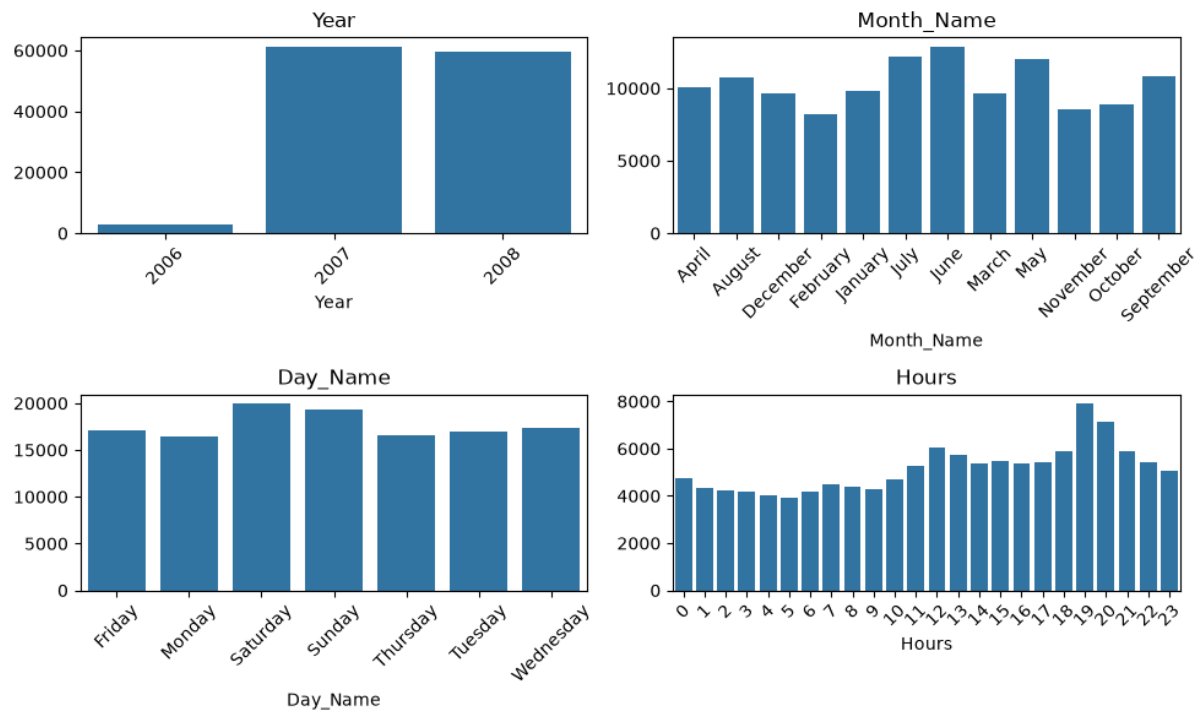


Figure 4. Global-reactive-power totals across year, month, day and hour categories.

Voltage

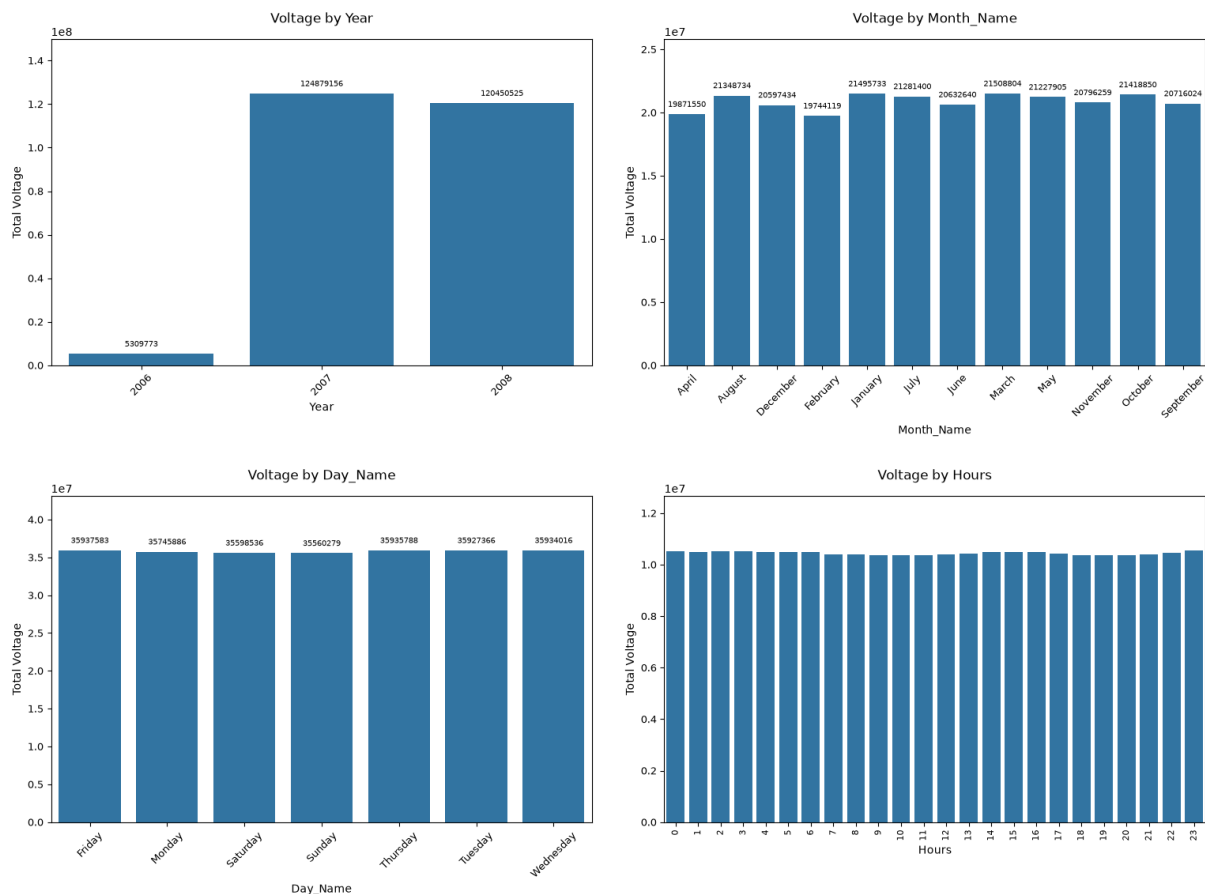


Figure 5. Voltage grouped analysis with value labels from the supplied notebook. The annual totals are much larger in 2007 and 2008 than in the partial 2006 period. Monthly, daily and hourly totals are comparatively close to one another, reflecting the relatively stable

voltage distribution observed in the histogram.

6. Key Findings

1. The analyzed dataset contains **1,044,506 records and 15 features**, providing a large volume of minute-level household electricity observations.
2. The notebook's **df.info()** output shows all 15 columns with 1,044,506 non-null values; no missing values are shown in the saved analysis.
3. The observations span **2006, 2007 and 2008**, with 2006 represented by a much shorter portion of the available period than the following years.
4. The **Global_active_power** distribution is strongly right-skewed: its mean is 1.1082 while its median is 0.550.
5. The **Global_reactive_power** distribution is also concentrated near low values, with a mean of 0.1183 and a maximum of 1.39.
6. The **Voltage** variable is comparatively stable, with a mean of about 239.96 V, median 240.21 V and standard deviation about 3.28 V.
7. The three sub-metering variables contain many zero-valued observations at their medians, while their maximum values are considerably larger than their typical values.
8. The grouped hourly charts show clear differences in electricity-related measurements throughout the day, making **hour** a useful candidate for future feature engineering.
9. The grouped analyses also show differences across **months and days of the week**, indicating that calendar features can be useful when modelling household consumption.
10. The exploratory distributions and grouped patterns provide a foundation for **future feature engineering, predictive modelling, anomaly investigation and deeper correlation analysis**.

7. Conclusion

The Household Electricity Consumption dataset provides a detailed minute-level view of electrical behaviour across multiple years. The analysis shows strong concentration and skewness in several consumption-related variables, while voltage remains comparatively stable. Time-related variables such as year, month, day and hour show meaningful variation and therefore provide useful dimensions for understanding electricity usage.

From a data-science perspective, the next stage can build on this EDA by creating time-based features, examining correlations between electrical measurements, investigating extreme observations, and developing forecasting or predictive models. The present report keeps the conclusions descriptive and grounded in the supplied notebook outputs rather than treating observed relationships as causal.

Technology used: Python | Pandas | NumPy | Matplotlib | Seaborn | Streamlit | Jupyter Notebook